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Systematic Mapping and Benchmarking of Sign Language Datasets

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1. Introduction

Sign languages are natural languages used by deaf and hard-of-hearing communities worldwide. In contrast to spoken languages, sign languages are visually conveyed through hand movements, facial expressions, and body posture, and have their own distinct linguistic structures. Recent advances in artificial intelligence and machine learning have increased research interest in automatic sign language recognition, translation, and accessibility technologies. Progress in these domains relies on the availability of suitable and well-documented datasets.

In the past decade, a wide range of sign language datasets have been introduced, covering various sign languages, recording environments, and research objectives. These datasets vary significantly in scale, modality (such as RGB video or depth data), annotation schemes, and signer diversity. However, information regarding these datasets is scattered across academic publications, project websites, and repositories. Consequently, researchers, particularly those new to the field, frequently encounter challenges in identifying suitable datasets and comprehending their limitations.

The lack of a consolidated and structured overview of sign language datasets hinders research progress and reproducibility. In the absence of clear comparisons and classifications, it is difficult to assess dataset usability, identify gaps in language coverage, or determine the suitability of datasets for specific research tasks. This challenge highlights the necessity for a systematic approach to mapping and evaluating existing sign language datasets.

This thesis aims to deliver a structured overview of existing sign language datasets by identifying, categorizing, and comparing their key characteristics. The analysis focuses on properties such as language coverage, modality, scale, annotation, and accessibility. By applying a systematic mapping methodology, the thesis clarifies the current landscape of sign language datasets and highlights existing limitations and research gaps.

2. Background and Related Work

2.1 Sign Language Datasets in Computational Research

Sign language datasets serve as the primary empirical foundation for computational sign language research. These datasets are essential for training, evaluating, and benchmarking models in tasks such as sign recognition, sign spotting, and sign language translation. Therefore, the design choices made during dataset construction directly affect the research questions that can be addressed and the validity of experimental results.

Existing datasets differ significantly in scope and objectives. Some are designed for isolated sign recognition, while others target continuous signing at the sentence or discourse level. Datasets also vary in signer diversity, annotation granularity, recording conditions, and modality coverage. These variations indicate differing research priorities and complicate the process of comparing results across datasets. Understanding the characteristics of available datasets is therefore essential for researchers when selecting appropriate resources or interpreting reported results.

2.2 Key Dimensions of Sign Language Dataset Design

Sign language datasets can be distinguished by several key dimensions. A primary distinction is the type of signing captured. Isolated sign datasets typically contain short clips where a single sign is performed, while continuous datasets include natural sign sequences with temporal dependencies. These two types of datasets support fundamentally different modeling approaches and evaluation strategies.

Another major dimension is data modality. Early datasets primarily relied on RGB video, whereas recent datasets incorporate additional modalities such as depth data, skeletal pose information, or multi-view recordings. Although multimodal datasets offer richer information, they also increase data complexity and hardware requirements.

Annotation practices further differentiate datasets. Some provide only gloss-level labels, while others include temporal segmentation, sentence alignment, or spoken-language translations. The annotation level determines whether a dataset is suitable for higher-level tasks such as translation or linguistic analysis.

Datasets also vary in recording conditions, from controlled studio environments to natural, in-the-wild settings. Factors such as background complexity, lighting variation, and camera setup influence dataset realism and model generalisability.

2.3 Dataset Scale, Diversity, and Accessibility

Scale is another defining characteristic of sign language datasets. Some contain a limited number of signs and signers, whereas others include hundreds of sign classes and tens of thousands of samples. Larger datasets typically enable more robust model training but require greater resources for collection and annotation. Signer diversity is a particularly important consideration. Datasets may include hearing signers, deaf signers, or both, and may vary in gender balance, handedness, and signing proficiency. Limited diversity can introduce bias and reduce the applicability of trained models in real-world contexts. Accessibility also varies among datasets. Many are publicly available, while others require special permission or are restricted to research institutions. Licensing terms, documentation quality, and long-term availability influence a dataset's practical usefulness and adoption within the research community.

2.4 Existing Reviews and Dataset Comparisons

Several surveys and review articles address sign language recognition and related tasks. These works typically focus on algorithmic developments, such as neural network architectures or feature extraction techniques. Datasets are generally presented as supporting components rather than as primary research objects.

Although some studies list commonly used datasets, they rarely offer a structured comparison of dataset characteristics. Information such as signer independence, annotation coverage, or licensing conditions is often scattered across individual papers and not systematically analysed. As a result, researchers seeking an overview of available datasets must manually inspect multiple sources, which is a time-consuming and error-prone process.

2.5 Need for a Systematic Mapping of Sign Language Datasets

The lack of a consolidated overview of sign language datasets highlights the need for a systematic mapping approach. Instead of evaluating model performance, this approach classifies datasets based on clearly defined attributes and identifies trends and gaps in dataset availability. Systematic mapping studies have been successfully applied in computer science research areas that contain many different types of studies and datasets and provide high-level insights into research landscapes. Applying this methodology to sign language datasets enables structured comparison across languages, dataset types, modalities, and evaluation practices. This perspective supports informed dataset selection, facilitates reproducibility, and guides future dataset development efforts.

3. Methodology

3.1 Research Design

A systematic mapping review methodology was utilized to provide a comprehensive overview of existing sign language datasets. This approach is particularly effective for identifying, categorizing, and highlighting trends and research gaps, with an emphasis on classification and comprehensive coverage of the research area. To clarify the scope of this study, it is important to distinguish between 'mapping,' which focuses on identifying broad patterns and trends, and 'synthesis,' which delves into detailed examination and analysis of specific topics. By adopting a mapping review, we can focus on broad patterns and trends in the field, aligning with Kitchenham's distinction between breadth-oriented mapping and depth-oriented synthesis, which reinforces the suitability of this approach for our study's objectives.

The methodology for this study follows established systematic review guidelines, specifically those proposed by Kitchenham et al. and Petersen et al. These frameworks guided the formulation of research questions, development of the research protocol, systematic search procedures, data extraction, and classification to ensure transparency and reproducibility.

3.2 Research Questions

The objective of this systematic mapping study is to identify and characterize existing sign language datasets and assess their usability and limitations. To achieve these aims, the following research questions were formulated:

- What sign language datasets have been published and are publicly accessible?
  Identifying these datasets benefits researchers and practitioners by providing a foundation for further research and helping dataset creators to build upon existing resources.
- What are the main characteristics of these datasets in terms of language coverage, modality, scale, and annotation?
  Understanding these characteristics helps practitioners designing datasets to meet specific research or application needs.
- What types of tasks (e.g., isolated sign recognition, continuous signing) do these datasets support?
  This question can guide model developers by outlining what tasks current datasets are suitable for, informing decisions about model design and task suitability.

- What gaps and limitations can be identified in current sign language datasets?
  By highlighting these gaps and limitations, dataset creators and researchers can identify opportunities for improvement and innovation, leading to the development of more comprehensive datasets and advancing the field.

These research questions are designed to facilitate the classification and mapping of datasets, rather than to conduct an in-depth empirical evaluation.

3.3 Review Protocol

A review protocol was established prior to the mapping study to ensure transparency and reproducibility. This protocol specifies the data sources, search strategy, selection criteria, and data extraction scheme used in the study.

3.3.1 Data Sources

The literature search utilized multiple academic databases and online repositories relevant to computer science and sign language research. Primary data sources included Google Scholar, IEEE Xplore, ACM Digital Library, and arXiv. Publicly available dataset repositories and project websites were also reviewed to identify datasets not indexed in traditional academic databases. While we aimed for comprehensive coverage, certain specialized resources like the Deaf Studies Digital Journal were not included due to the scope trade-offs, focusing on datasets with a more technological orientation. This selection strategy ensures relevance to the primary domains of interest but acknowledges potential limitations in interdisciplinary coverage.

3.3.2 Search Strategy

A keyword-based search strategy was implemented to identify publications describing sign language datasets. Search terms combined concepts related to sign language and datasets, including *sign language*, *dataset*, *corpus*, and *recognition*. Boolean operators were used to refine search results and exclude irrelevant records. To enable reproducibility, a representative Boolean search string used in the process was: "sign language AND dataset AND (corpus OR recognition) AND NOT gesture". This provides a snapshot of the approach and allows readers to replicate the search process.

The search process was iterative. Initial searches were supplemented by backward snowballing, in which references from relevant papers were examined to identify additional datasets. The snowballing process was carefully monitored and terminated upon reaching a saturation threshold, where no new relevant datasets were discovered in five consecutive papers. This choice of 'five consecutive papers' was informed by the principle of diminishing returns, where the likelihood of finding new, relevant datasets decreases significantly with each paper analyzed, providing a practical and defensible stopping point. To ensure comprehensive coverage of publicly documented datasets, manual searches of dataset project websites were also conducted.

3.3.3 Inclusion and Exclusion Criteria

To ensure consistency in dataset selection, explicit inclusion and exclusion criteria were defined and applied during the screening process.

Studies and datasets were included if they:

- described a dataset related to a natural sign language,
- provided visual data (video or image-based),
- were documented in a peer-reviewed paper or an official dataset description,
- were written in English.

Studies and datasets were excluded if they:

- focused on gesture recognition unrelated to sign languages,
- lacked sufficient documentation,
- described synthetic or simulated data only,
- were not accessible or identifiable through public documentation.

3.3.4 Study Selection Process

The study selection process was conducted in two stages. First, titles and abstracts were screened to exclude studies that were clearly irrelevant. Second, the full texts of the remaining publications were reviewed to determine eligibility according to the inclusion and exclusion criteria. The final set of datasets was selected after full-text assessment.

A structured data extraction scheme was developed to systematically capture relevant characteristics of each dataset. Extracted information was recorded in a spreadsheet to ensure consistency and facilitate comparison across datasets. This approach not only supports the current analysis but also lays the groundwork for future benchmark goals by enabling a detailed assessment of dataset attributes that can contribute to more targeted benchmarking analyses.

The extracted attributes include, but are not limited to:

- dataset name and publication details,
- sign language and country or region,
- dataset scale (number of signs, samples, and signers),
- dataset type (e.g., isolated signs, continuous signing),
- modalities provided (e.g., RGB video, depth data, skeleton data),
- annotation type and availability of benchmark results,
- accessibility and licensing information.

This extraction scheme forms the basis for the classification and benchmarking conducted in this study.

3.5 Classification and Mapping

Based on the extracted attributes, datasets were classified along multiple dimensions, including language coverage, modality, dataset type, and annotation characteristics. These classification dimensions establish a taxonomy that enables comparison between datasets and supports the identification of research trends and gaps. The mapping results are presented using descriptive tables and summaries, rather than statistical synthesis, in accordance with the objectives of a systematic mapping review.

3.6 Validity Considerations

As with any systematic study, this mapping review is subject to potential threats to validity. These include the possibility of missing relevant datasets due to limitations of the selected databases or search terms, as well as subjective judgment during dataset selection and classification. To mitigate these threats, multiple data sources were used, an iterative search strategy was applied, and clearly defined selection criteria guided the screening process.

4. Results and Initial Discussion

4.1 Overview of Included Datasets

A total of seven sign language datasets were identified through the systematic search and selection process outlined in Chapter 3. These datasets were chosen to ensure coverage across various sign languages, dataset sizes, and design features. The selected datasets cover sign languages from several geographic regions and differ in dataset type, modality, annotation detail, and accessibility. Each dataset is described in peer-reviewed publications and includes sufficient metadata for structured analysis. This chapter reports the results of dataset characterization according to the predefined extraction criteria. The findings are structured around key dimensions relevant to dataset usability, accessibility, and research applicability.

4.2 Distribution by Sign Language and Region

The seven datasets cover a diverse range of sign languages, including national sign languages from Europe, South America, Asia, and North America. Each dataset targets a single sign language, reflecting the language-specific approach to sign language data collection. Despite broad geographic representation, the distribution is uneven. Most originate from regions with established research infrastructure in computer vision and machine learning. Several regions and sign languages are underrepresented, indicating gaps in dataset availability. These findings highlight the fragmented development of sign language datasets, which are typically created within isolated national or institutional contexts rather than through coordinated international initiatives.

4.3 Dataset Type and Task Support

Analysis of dataset types indicates that most included datasets focus on isolated sign recognition**.** These datasets generally consist of short video clips, each representing a single sign. Few datasets support continuous signing**,** which involves sequences of signs forming sentences or phrases. Continuous datasets are more complex to collect and annotate, which may explain their lower prevalence. As a result, current publicly available datasets are more suitable for isolated sign recognition tasks than for advanced applications such as sign language translation or discourse-level analysis.

4.4 Dataset Scale and Composition

The included datasets vary substantially in scale. Some contain a limited number of sign classes and samples, while others include hundreds of signs and tens of thousands of video instances. The number of signers also varies considerably across datasets. Larger datasets typically include more signers, supporting greater variation in signing style and appearance. However, detailed demographic information, such as age range or linguistic background, is frequently absent or inconsistently reported. Variation in scale and documentation affects direct comparison between datasets and restricts the assessment of signer diversity across studies.

4.5 Modalities and Recording Setup

Most datasets utilize RGB video as the primary modality. Several datasets also provide additional modalities, such as depth data or skeletal pose information, typically captured with depth sensors. Multimodal datasets provide richer representations but are less common and generally require specialized hardware. In contrast, RGB-only datasets are more prevalent and easier to collect, though they may offer limited detail for fine-grained sign analysis. Recording environments range from controlled studio settings with simple backgrounds to more complex environments with varied backgrounds and lighting. Controlled setups are predominant in isolated-sign datasets, whereas continuous datasets typically exhibit greater environmental variability.

4.6 Annotation and Benchmarking Practices

All included datasets offer some form of annotation, most commonly gloss-level labels for individual signs. However, annotation depth and structure differ considerably across datasets. Only a subset of datasets explicitly defines signer-independent evaluation protocols, in which test data consists of signers not present during training. These protocols are essential for evaluating model generalizability but are not consistently implemented. Not all datasets provide baseline model results. When included, baseline evaluations vary in model architecture, evaluation metrics, and reporting standards, which limits direct performance comparisons across datasets.

4.7 Accessibility and Availability

All datasets included in this initial analysis are described as publicly available or accessible upon request for research purposes. However, licensing information is often incomplete or vaguely specified. Documentation quality also varies. Some datasets include comprehensive descriptions, download instructions, and evaluation protocols, whereas others provide minimal guidance beyond the associated publication. These differences affect dataset accessibility and practical usability, especially for researchers who are new to the field.

4.8 Initial Discussion

The initial mapping indicates a strong emphasis on isolated sign recognition datasets and controlled recording environments. While this supports early-stage model development, it restricts the evaluation of model generalization and real-world applicability. The lack of consistent signer-independent benchmarks and incomplete demographic reporting further complicates cross-dataset comparisons. These findings indicate a need for improved dataset documentation practices and standardized evaluation criteria, which will be explored further as the dataset corpus expands.

5. Conclusion

This thesis presents a systematic mapping and initial benchmarking of sign language datasets, offering a structured overview of their key characteristics and current limitations. By identifying and categorizing datasets based on language coverage, modality, scale, and annotation, this work establishes a clear baseline for understanding the current landscape of sign language research datasets. The initial findings indicate that many available datasets primarily support isolated sign recognition and are collected in controlled environments. Although these datasets are valuable for early-stage research, they provide limited support for evaluating generalization and real-world applicability. Furthermore, inconsistencies in dataset documentation and evaluation practices hinder direct comparison across datasets. The mapping and benchmarking framework proposed in this thesis serves as a foundation that can be extended as additional datasets are incorporated. Future research can build on this baseline by expanding the dataset corpus, refining evaluation criteria, and examining inclusivity and accessibility in greater detail. Overall, this study provides a transparent starting point for more systematic analysis and informed utilization of sign language datasets.

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